Chapter

10 Science inquiry skills

- plan and conduct investigations
- analyse data, draw conclusions, evaluate investigation design and findings
- evaluate the impact of advancements in human biology on individuals and society
- communicate understandings of human biology.

MULTIPLE-CHOICE QUESTIONS

The next two questions refer to the information below.

For a number of years, there has been a belief among some people that the immunisation of infants and children causes autism. Many studies have focused on potential links between the ingredients in vaccines and their possible side effects. Mercury has been one of the ingredients tested. Numerous long-term experiments have been conducted to determine whether there is a link between autism and the presence of mercury in vaccines. However, no link has been established.

1. *(2015:1.21)*

Which of the following is a logical hypothesis for such an experiment?

(a) Autism is caused by mercury.

(b) Vaccines without mercury are better for infants and children.

(c) Vaccines containing mercury do not cause autism.

(d) Do vaccines containing mercury cause autism?

2. *(2015:1.22)*

The dependent variable in this experiment would be the

(a) volume of vaccine administered.

(b) number of infants and children in the study.

(c) amount of mercury in the vaccine.

(d) number of new cases of autism

3. *(2017:1.04)*

A researcher recorded the body temperature of seven participants during an investigation into the effects of exercise. The results are shown below.

37.4, 37.9, 36.8, 37.2, 38.0, 37.2, 37.5

Which of the following calculates correctly the mean, median and range for the results?

	Mean	Median	Range
(a)	37.5	37.2	1.1
(b)	37.4	37.4	1.2
(c)	37.2	37.5	1.0
(d)	37.4	37.2	1.2

4. *(2017:1.15)*

Which of the following is the **best** way of improving the reliability of an investigation?

(a) ensure correct calibration of equipment

(b) incorporate more controlled variables into the experimental design

(c) repeat the entire investigation

(d) make the investigation into a double-blind study

5. *(2018:1.09)*

For which of the following sets of data is a line graph not suitable?

(a) the body mass of all female students at a high school

(b) the average body fat content of a particular teenage boy at different ages

(c) the average pulse rate of a student before, during and after exercise

(d) the average amount of food eaten by teenage boys at different environmental temperatures

The next two questions refer to the information and table below:

Lydia, Jake and Ahmed conducted an experiment to investigate the effects of drinks containing caffeine on urine output.

Jake drank nothing for two hours and then drank one litre of water as quickly as he could. His urine was collected every 20 minutes for the following two hours. He again drank nothing for two hours before repeating the test with Coke Zero® and then with Red Bull Sugarfree®. The results are recorded in the table below.

Time (minutes)	Urine output (mL)		
	Water	Coke Zero®	Red Bull Sugarfree®
20	60	63	61
40	81	90	97
60	75	85	95
80	52	65	71
100	41	43	50
120	43	45	42

6. *(2018:1.29)*

From the data in the table, which of the following statements could be correct?

(a) drinking water causes urine output to decrease

(b) drinking Coke Zero® has no effect on urine output

(c) drinking Red Bull Sugarfree® causes urine output to increase

(d) drinking water causes a greater increase in urine output than Coke Zero®

7. *(2018:1.30)*

From the data in the table, which of the following statements could be correct?

(a) The experiment is not valid because it does not have a control but is reliable due to the repeated trials.

(b) The experiment is not valid because it does have a control and is not reliable due to having repeated trials.

(c) The experiment is valid due to variables being kept the same and reliable because of repeated trials.

(d) The experiment is valid due to variables being kept the same but not reliable due to there being no repeated trials.

SHORT ANSWER QUESTIONS

8. [13 marks] *(2013:2.36)*

A recent study investigated the effects of MDMA (ecstasy) on human thermoregulation. The research team selected 10 individuals (male and female) aged between 18 and 35 years. Each participant attended two sessions, one week apart. At the first session participants received, in tablet form, either a placebo or 2 mg/kg of MDMA. If at the first session participants received a placebo, at the second session they received MDMA, and vice versa.

At each session, participants assembled in a room at 10 am with the room temperature at 23°C. After 30 minutes, the room temperature was changed to 30°C, which took about 2 minutes. Data collection began at 11 am, when the drug or placebo tablet was given with a small amount of water. Recordings of core body temperature were taken every hour for the next 4 hours, with the session concluding at 3 pm.

Data for each of the **five** time periods were averaged and recorded (in order of time starting at 11 am) in a notebook as follows:

MDMA: 36.9°C, 37.1°C, 37.5°C, 37.6°C, 37.6°C
Placebo: 36.9°C, 37.0°C, 37.0°C, 37.1°C, 37.1°C

(a) Present the above data in a table. [5]

(b) Formulate a hypothesis for this experiment. [1]

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(c) Explain why each participant did not receive the same amount of MDMA: that is, they received 2 mg of MDMA per kilogram of body mass of the participant. [1]

(d) (i) Describe **two** variables that were controlled adequately in the experiment. [2]

(ii) For **one** of the variables described in part (d) (i), explain why it needed to be controlled. [1]

(e) The research team also recorded the oxygen consumption of participants over the same time period. While they found that oxygen consumption after administration of the placebo remained constant, they found that it increased significantly after the administration of MDMA.

Using this information and data from the table, suggest what caused the observed effect of MDMA on core body temperature. [1]

(f) (i) During both sessions, the research team took recordings of skin temperature for 5 hours from 10 am. Between 10.30 am and 11 am, the skin temperature increased 1°C prior to administration of the placebo and MDMA. Explain why this occurred. [1]

(ii) After 12.30 pm, the skin temperature, following administration of MDMA, steadied at 0.5°C above the skin temperature following administration of the placebo. Account for what might have caused the difference in skin temperatures between the two treatments. [1]

9. [10 marks] *(2014:2.35)*

An experiment was conducted on the effects of fluid consumption on urine production. The experiment involved the comparison of water consumption with the consumption of saline solution. Saline solution is a sterile solution of water and salt (normally sodium chloride). The experiment involved 30 subjects, 15 who consumed one litre of water in a five minute period and 15 who consumed one litre of saline solution in the same five minute period. All subjects were required to stay in a small room maintained at a temperature of 25°C and were asked to perform minimal physical activity. Urine production over the three hours following fluid consumption was recorded for all subjects. The results for each group were averaged and are presented below.

Time (minutes)	**Urine production (mL)**	
	Water consumption	Saline solution consumption
0	24	18
30	360	21
60	450	27
90	255	36
120	48	29
150	30	34
180	27	24

(a) (i) Propose a hypothesis for the experiment. [1]

(ii) List **two** variables that were controlled in the experiment. [2]

(b) Suggest how researchers could increase the

(i) validity of the experiment. [1]

(ii) reliability of the results. [1]

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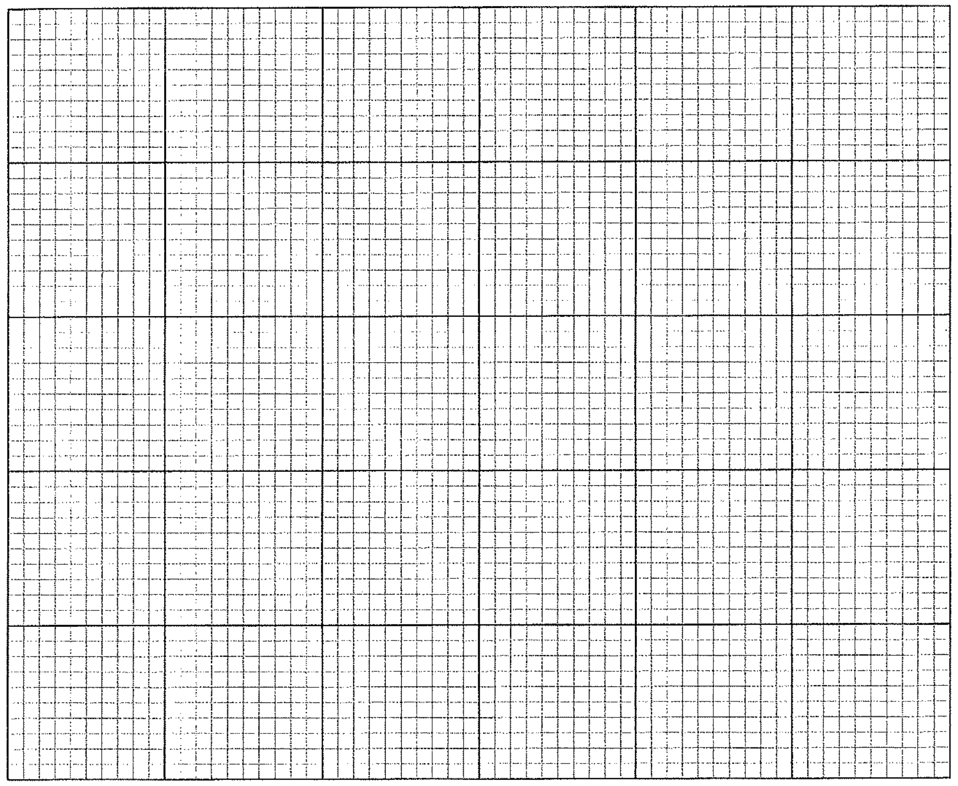
(c) Graph the data in the table on the grid below. [5]



10. [9 marks] *(2015:2.36)*

Apnoea of prematurity is a common condition in premature infants (infants born earlier than 38 weeks gestation). Apnoea results in a complete stoppage in breathing for up to 20 seconds or more. Apnoea has various causes but if left untreated can result in low blood pressure and heart rate, brain damage, and sometimes death. Infants with apnoea of prematurity often require help with their breathing, which may include insertion of a breathing tube (intubation) or being supplied with oxygen-rich air in the humidicrib. One of the most common drug treatments for apnoea of prematurity is daily doses of caffeine, which helps to stimulate the breathing reflex and eliminate the symptoms of apnoea.

Presented below are some of the results from an experiment on the effects of caffeine in premature infants. In group A, 960 infants were administered daily doses of oral caffeine. In group B, 932 infants were administered daily doses of an oral placebo. All infants in the experiment were born between 26 and 28 weeks gestational age. All were given daily doses of either caffeine or the placebo for six weeks.

	Mean age of infants (gestational age) in weeks	
	Group A	**Group B**
Age at first dose	27.8	28.1
Age at last dose	33.8	34.1
Age at last use of breathing tube (intubation) required	29.5	30.4
Age at last use of additional oxygen required	33.7	35.3

(a) Identify **two** variables that were controlled in the experiment. [2]

(b) During the experiment, group B received a placebo.

(i) Define 'placebo'. [1]

(ii) Why are placebos usually administered in experiments? [1]

(c) With reference to the data provided, state **one** conclusion that can be drawn from the experiment. [1]

Additional data collected during the study indicated that group A infants gained less weight than group B infants.

(d) Given this additional data, formulate a new hypothesis that researchers could now be interested in studying. [1]

(e) (i) A change in concentration of a particular compound is the most important factor in stimulating the breathing reflex. Name this compound. [1]

(ii) Breathing can be altered voluntarily, allowing individuals to increase or decrease breathing rate or even stop breathing for a short time. What part of the brain voluntarily controls breathing? [1]

(iii) Why is it important for humans to have voluntary control over breathing rate? [1]

11. **[5 marks]** *(2016:2.33)*

A study has recently been carried out to find a way by which Type 1 diabetics may be able to eliminate daily insulin injections. Scientists have investigated the possibility of the use of a particular type of therapy to control blood glucose levels and regulate glucose metabolism.

A small sequence of DNA was injected into the veins of diabetic rats, creating cells that produce insulin. The scientists found that, for up to six weeks, a single injection enabled the diabetic rats to show the same control of glucose levels as that found in healthy rats.

(c) Describe a control group that could be used for this experiment. [2]

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(d) State **one** factor involving blood glucose levels in rats that the scientists would have to determine before they began the experiment. [1]

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(f) List **two** further steps that scientists would have to take before people with Type 1 diabetes would be able to have access to this therapy. [2]

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12. **[6 marks]** *(2016:2.36)*

Use the table below to complete the following questions.

Time (mins)	Level of anti-diuretic hormone (ADH) in the bloodstream (arbitrary units)	Rate of urine production in the kidney (mL/min)
0	7.5	2.1
2	7.5	2.1
4	7.0	2.1
6	17.0	1.5
8	22.0	1.3
10	7.0	1.5
12	7.5	2.1
14	7.5	2.1

(a) Graph these results on the grid provided below. [5]

(b) From the graph, determine the level of ADH in the bloodstream after seven minutes. [1]

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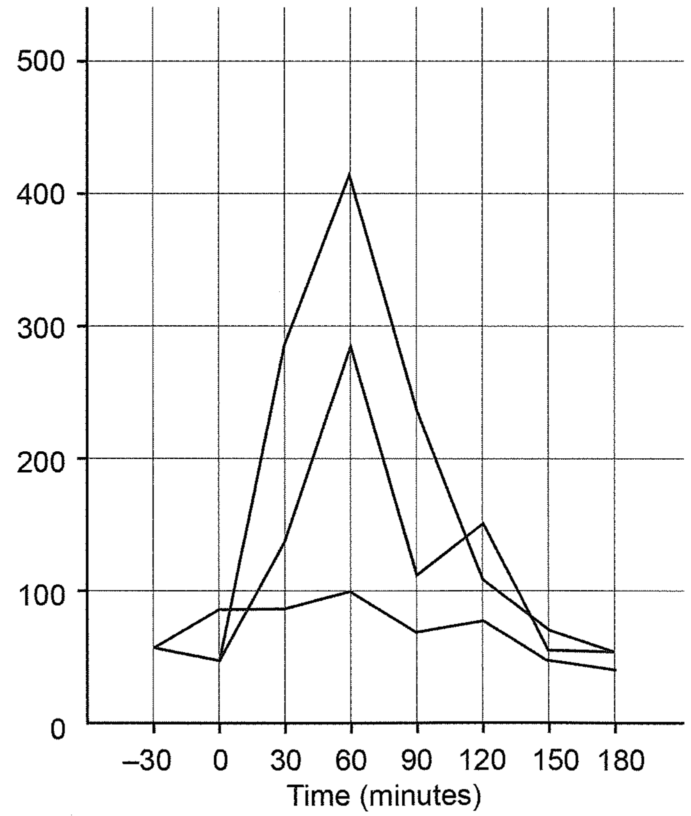
14. [11 marks] *(2017:2.33)*

Scientists decided to investigate how urine production was affected by drinking different salt solutions. They chose three groups of 10 volunteers. Each group drank 1 litre of different solutions: Group 1 distilled water; Group 2 10% salt solution; and Group 3 30% salt solution. Urine samples were collected from each person that participated in the experiment 30 minutes before they drank the solution and then every 30 minutes afterward, and the volume of urine was then recorded. The graph and table below show the average volume of urine collected every 30 minutes from each group.

	Average urine production (mL)		
Time (minutes)	**Group 1 Distilled water**	**Group 2 10% Salt solution**	**Group 3 30% Salt solution**
-30	56	58	55
0	48	47	51
30	287	138	85
60	415	285	98
90	235	112	67
120	103	82	77
150	68	54	46
180	54	52	40

The graph shown below is a student's attempt to represent the data shown in the table.

Effect on urine production



(a) Identify **three** errors in the graph. [3]

(b) Why was urine collected from each volunteer 30 minutes prior to their drinking 1 litre of their solution? [2]

(c) Identify **two** variables **not** indicated in the information for this question that needed to be maintained across all three groups during the investigation to ensure a fair test was conducted. [2]

(d) State **one** factor that is changed in the normal internal environment due to the drinking of 1 litre of 30% salt solution. [1]

(e) At what time would the blood concentration of ADH be the lowest for the people in Group 1? Justify your answer. [3]